



Koneru Lakshmaiah Education Foundation

(Deemed to be University estd. u/s. 3 of the UGC Act, 1956)

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DATABASE SYSTEMS ENGINEERING AND DISTRIBUTED BACKEND DEVELOPMENT (25CS1302E)

Y25 - 2026-2027 Trimester - 4

Project Abstract Submission Form

Sec No-2	Team No-9	Project Title-Library Management with RFID Integration
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1. Abstract

The Library Management with RFID Integration project aims to develop an automated and efficient system for managing library books and user transactions using Radio Frequency Identification (RFID) technology. Traditional library management systems often depend on manual data entry and barcode scanning, which can be time-consuming and may lead to errors. This project addresses these limitations by integrating RFID technology into the process of book identification, issue, return, and inventory management. In the proposed system, each book is equipped with an RFID tag containing a unique identification number. RFID readers are used to automatically detect and identify books without requiring direct contact or line-of-sight scanning. Students or library members can use RFID-enabled identification cards to perform borrowing and returning transactions quickly. The system stores important information such as book details, user details, issue dates, return dates, and availability status in a centralized database. The system also helps librarians monitor book inventory and locate missing or misplaced books more efficiently.

Team Details

Roll Number(s)	Name of the Student	Signature
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1. Frameworks / Platforms / Software Used

Arduino / ESP32 – Used as the main controller for RFID communication and system operations.

RFID Module (RC522) – Used to read RFID tags attached to books and user ID cards.

Embedded C / C++ – Used to develop the microcontroller program.

MySQL – Used to store and manage book, student, and transaction records.

Python / Flask – Can be used to develop the web-based backend and connect the RFID system with the database.

Guide Name and Signature (remarks if any)